

A Dual Operator for Prime–Zero Coupling

and a Conditional Proof of Energy Asymmetry

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Abstract

We introduce the operator $\tilde{T} := \Phi \circ \Phi^*$, the dual of the loop operator $T = \Phi^* \circ \Phi$ constructed in Paper 3 of this series. Here $\Phi: H_{\text{str}} \rightarrow H_{\text{null}}$ maps a finite-dimensional prime space into a finite-dimensional zero space via Gaussian-damped prime resonance vectors.

What is proved. We establish (§2) the algebraic spectral identity $\sigma(T) \setminus \{0\} = \sigma(\tilde{T}) \setminus \{0\}$ by classical operator theory; the maximum discrepancy between shared eigenvalues is below 10^{-15} (machine precision). We prove (§3.4) that $W_1 := C_T \cdot \tilde{T}^+$ is self-adjoint. W_1 does not assume RH or the HP postulate; however, its normalization constant C_T is calibrated against the zero ordinates via OLS and should not be presented as fully input-free. $\tilde{T}$ is explicit and deterministic; the γ_k are external numerical inputs, not conclusions of the model (§3.4). We prove (§4) the Abel Summation Principle (Lemma M3): for monotone decreasing weights f_i and a sequence g_i with bounded partial sums, $|\sum_i f_i g_i| \leq M f_1$. We prove (§4) the Prime Exponential Sum Bound (Theorem EXPLICIT): for fixed $\gamma > 0$,

$$M_k(\kappa) := \max_{\substack{x \leq \kappa \\ p \leq x}} \left| \sum \sin(\gamma_k \log p) \right| = O\left(\frac{\pi(\kappa)}{\gamma_k}\right)$$

using only the Prime Number Theorem (no Riemann Hypothesis). We prove (§2) that $\tilde{T}$ is not a Hilbert–Pólya operator: the correlation $r_2 = \text{corr}(\omega_j, \gamma_{k(j)})$ falls from 0.50 to 0.16 for $\kappa \in \{23, \dots, 503\}$.

What is numerical. The eigenvectors of $\tilde{T}$ localize at zero ordinates γ_k with degrees 0.45–0.99 [Numerical, §3.2]. The correlation $r_1 = \text{corr}(\mu_j, 1/\gamma_{k(j)}) = 0.950$ ($\kappa = 53$) [Numerical, §3.1]. The energy asymmetry $\eta_{\text{orig}} > 0$ holds for all tested $\kappa \leq 1009$ [Numerical, §5.1].

What is conditional. Under the Assumption (Weyl equidistribution of prime log-phases, a condition on $\text{Re}(s) = 1$, distinct from the Riemann Hypothesis) we prove (§5.3) that the normalised cross/stream averages vanish [conditional]. We establish numerically that $\Delta(\kappa) \geq \Delta_{\text{Burst}} \approx 3.11 > 0$ for all tested $\kappa > 7$ [numerical lower bound, κ -invariant] and $\eta_{\text{orig}}(\kappa) \rightarrow \eta_{\infty} \approx 0.81 > 0$ [numerical].

What remains open. The analytic proof of $\eta_{\text{orig}} > 0$ without the above assumption (§6); whether a circularity-free spectral function $f(\tilde{T})$ exists with spectrum converging to $\{\gamma_k\}$ (§6); and the arithmetic origin of $C_{\eta} \approx 0.39$ (§6).

Structure. The formal core in §§2–5 is non-circular and proof-complete. Section 6 states the open problems. (clearly marked, not used in proofs). No use of the Riemann Hypothesis, the GUE conjecture, the Montgomery pair correlation conjecture, or the Hilbert–Pólya postulate is made anywhere in §§2–5.

Notation and Conventions

Critical line. Throughout this series, the *critical line* means $\Re(s) = \frac{1}{2}$, parameterised by $s = \frac{1}{2} + i\gamma$ with $\gamma \in \mathbb{R}$.

Global parameters. All numerical results use the normative parameter set: prime cutoff $\kappa = 53$ ($\pi(\kappa) = 16$ active primes), Gaussian regularisation parameter $\varepsilon = 0.05$, $N = 100$ Riemann zero ordinates, and evaluation point $\sigma_0 = \frac{1}{2}$.

Main objects. The encoding map $\Phi: H_{\text{str}} \rightarrow H_{\text{null}}$ assigns to each prime $p \leq \kappa$ a resonance vector $a_p = (e^{-\varepsilon^2 \gamma_k^2 / 2} \sin(\gamma_k \log p))_{k=1}^N$. The loop operator $T = \Phi^* \circ \Phi$ acts on the prime space H_{str} ; its dual, the Tehrani operator $\tilde{T} = \Phi \circ \Phi^*$, acts on the zero space H_{null} and is the central object of this paper. The energy asymmetry $\eta_{\text{orig}} = 1 - c^T G^{\text{un}} c / E_{\text{str}}$ measures the gap between structural and spectral energy. Two arithmetic constants govern spectral scaling: $C_\eta \approx 0.39$ (renormalised eigenvalue growth) and C_T (OLS slope of μ_j vs. $1/\gamma_{k(j)}$).

Not to be confused with. $C_\eta \neq C_T$: C_η governs $\lambda_{\text{max,ren}} \sim C_\eta \cdot \pi(\kappa)$, while C_T governs $\mu_j \sim C_T / \gamma_{k(j)}$. The inverse operator $W_1 = C_T \cdot \tilde{T}^+$ is *not* a Hilbert–Pólya operator; its HP-correlation r_2 decays to 0.16 with growing κ .

1 Introduction

1.1 Background and Motivation

This paper is the fourth in a series developing an operator-theoretic framework for studying the distribution of the non-trivial zeros of the Riemann zeta function. The formal content in §§ 2–5 is non-circular and proof-complete.

Paper 1 [9] introduced the local curvature function and established the divergence $H_{\text{local}}(\frac{1}{2}, \kappa) \sim 2(\log \kappa)^2 \rightarrow \infty$ at $\sigma = \frac{1}{2}$.

Paper 2 [10] connected this local curvature to the Weil explicit functional, constructing admissible test functions for which the curvature divergence becomes a structural feature of the Weil energy.

Paper 3 [11] made the geometric structure explicit: two finite-dimensional real Hilbert spaces H_{str} and H_{null} connected by a linear map Φ , with loop operator $T = \Phi^* \circ \Phi$ and energy asymmetry $\eta_{\text{orig}} \in [0.598, 0.704]$ (numerical). These results appear in § 1 for context; the formal core §§ 2–5 depends only on the definitions in § 2.1.

The present paper introduces and analyzes $\tilde{T} = \Phi \circ \Phi^*$, the dual of T , acting on the zero space H_{null} . The operator emerges canonically from the same Φ and requires no new postulate. Several established programs seek a spectral interpretation of the Riemann zeros; we position the present work relative to the main ones.

A Hilbert–Pólya operator $\mathcal{H}$ would satisfy $\mathcal{H}\psi_k = \gamma_k\psi_k$; the zero ordinates appear directly as eigenvalues. The Tehrani operator $\tilde{T}$ has eigenvalues $\mu_j \sim C_T/\gamma_{k(j)}$ (numerically); the zero ordinates appear as localization centers of eigenvectors, not as eigenvalues. We use the standard formulation of the Hilbert–Pólya setting (see e.g. the survey by Bombieri [2]) without adopting it as a postulate. We call $\tilde{T}$ a “resonance operator” to express eigenvector concentration around arithmetic centers γ_k ; this is paper-specific terminology, not used in the scattering-theoretic sense.

Berry and Keating [1] seek a self-adjoint operator whose eigenvalues are γ_k via xp -quantisation; the present paper operates in the opposite direction (γ_k as eigenvector centers, not eigenvalues). Connes [4] gives an adelic trace-formula interpretation; no formal connection to $\tilde{T}$ is established here. Sierra and Townsend [7] provide a truncated Hamiltonian realization connected to Connes’ absorption spectrum; $\tilde{T}$ is not a Hamiltonian but a dual encoding operator. The Bost–Connes system [3] operates in a quantum-statistical framework with KMS states; the present construction is purely finite-dimensional and linear-algebraic. Goldston and Gonek [5] provide an independent confirmation of $M_k(\kappa) = O(\pi(\kappa)/\gamma_k)$ via mean-value methods. The closest historical precedent for the oscillatory-sum estimate is Turán’s [8] program on approximate Dirichlet polynomials; the novelty of our Theorem EXPLICIT lies in how it enters the conditional energy-asymmetry argument of § 5.

The main conditional result is a proof that $\eta_{\text{orig}} > 0$ under the Assumption — Weyl equidistribution of prime log-phases, a condition on $\text{Re}(s) = 1$, distinct from RH which concerns

$$\operatorname{Re}(s) = \frac{1}{2}.$$

1.2 Main Results

We prove the algebraic spectral identity $\sigma(T) \setminus \{0\} = \sigma(\tilde{T}) \setminus \{0\}$ (S2), self-adjointness of $W_1 = C_T \cdot \tilde{T}^+$ (§3.4), the Abel Summation Principle (Lemma M3, §4.3), and the Prime Exponential Sum Bound $M_k(\kappa) = O(\pi(\kappa)/\gamma_k)$ using only the Prime Number Theorem (§4.4). We establish numerically that $\tilde{T}$ is not a Hilbert–Pólya operator (§2.4), that $\eta_{\text{orig}} > 0$ for all tested $\kappa \leq 1009$ (§5.1), and that the κ -invariant lower bound $\Delta(\kappa) \geq \Delta_{\text{Burst}} \approx 3.11 > 0$ holds (§5.2). Under the Assumption, the normalised cross-term averages vanish (§5.3).

1.3 Reader Contract

This paper *proves*, by classical operator theory and the Prime Number Theorem, the spectral identity $\sigma(T) \setminus \{0\} = \sigma(\tilde{T}) \setminus \{0\}$, self-adjointness of W_1 , the Abel Summation Principle, and the bound $M_k(\kappa) = O(\pi(\kappa)/\gamma_k)$. These are unconditional.

This paper *establishes numerically* that $\tilde{T}$ is not a Hilbert–Pólya operator ($r_2 \rightarrow 0.16$), that $\eta_{\text{orig}} > 0$ for $\kappa \leq 1009$, and that $\Delta_{\text{Burst}} \approx 3.11 > 0$ is κ -invariant. These are verified on a finite grid and labelled [numerical].

This paper *states conditionally*, under the Assumption (Weyl equidistribution on $\operatorname{Re}(s) = 1$), that the normalised cross/stream averages vanish.

This paper *leaves open* six problems: the analytic proof of $\eta_{\text{orig}} > 0$ (Problem 6.2), the equidistribution condition (Problem 6.1), a circularity-free HP spectral function (Problem 6.3), the arithmetic origin of $C_\eta \approx 0.39$ (Problem 6.4), the asymptotics of $C_T(\kappa)/\pi(\kappa)$ (Problem 6.5), and whether $\eta_{\text{orig}} > 0$ follows from PNT alone (Problem 6.6).

2 The Tehrani Operator

2.1 Recollection: H_{str} , H_{null} , Φ , T

We fix throughout:

- A prime cutoff $\kappa > 1$, with active prime set $S(\kappa) := \{p \text{ prime} : p \leq \kappa\}$ and $\pi(\kappa) := |S(\kappa)|$.
- The first N positive imaginary parts of nontrivial zeros of $\zeta(s)$, ordered $0 < \gamma_1 \leq \gamma_2 \leq \dots \leq \gamma_N$.
- A Gaussian damping parameter $\varepsilon > 0$.

Definition 2.1 (H_{str} and H_{null} , from Paper 3).

$$H_{\text{str}} := \ell^2(S(\kappa)), \quad \dim H_{\text{str}} = \pi(\kappa). \quad H_{\text{null}} := \ell^2(\{\gamma_k\}_{k=1}^N), \quad \dim H_{\text{null}} = N.$$

Both spaces carry standard inner products. We work throughout with real vectors and real operators.

Definition 2.2 (Resonance vectors, from Paper 3). *For each prime $p \in S(\kappa)$, the resonance vector $a_p \in H_{\text{null}}$ is defined by*

$$a_p := \left(e^{-\varepsilon^2 \gamma_k^2 / 2} \sin(\gamma_k \log p) \right)_{k=1}^N.$$

The Gaussian envelope ensures $a_p \in \ell^2(\{\gamma_k\})$ uniformly in κ .

Definition 2.3 (Linear map Φ , from Paper 3).

$$\Phi: H_{\text{str}} \rightarrow H_{\text{null}}, \quad \Phi(e_p) = a_p, \quad \Phi(c) = \sum_{p \leq \kappa} c_p a_p.$$

Definition 2.4 (Loop operator T , from Paper 3).

$$T := \Phi^* \circ \Phi: H_{\text{str}} \rightarrow H_{\text{str}}, \quad T_{pq} = \langle a_p, a_q \rangle_{H_{\text{null}}} = G_{pq}^{\text{un}}.$$

T is self-adjoint, positive semi-definite, and identical to the Gram matrix G^{un} of the resonance vectors.

Definition 2.5 (Canonical weight vector, from Paper 3). *The normative weight vector $c \in H_{\text{str}}$ has components $c_p := \sqrt{V_p(\frac{1}{2})}$, where $V_p(\sigma) = \frac{d^2}{d\sigma^2} \log \|M_\sigma f_{\kappa,p}\|^2$. The exact formula is $c_p = \sqrt{4(\log p)^2(2p-1)/(p(p-1)^2)}$. The leading-term ($K=1$) approximation $c_p \approx 4 \log(p)/\sqrt{p}$ is asymptotically close but numerically differs for small primes; the exact formula is used throughout this paper and in all verification scripts.*

Reference values ($\varepsilon = 0.05$, $\sigma = \frac{1}{2}$): $c_2 \approx 1.698$, $c_3 \approx 1.418$, $c_5 \approx 1.080$, $c_7 \approx 0.884$.

Definition 2.6 (Energy quantities, from Paper 3).

$$E_{\text{str}} := \sum_{p \leq \kappa} c_p^2 \|a_p\|^2 = c^T D c, \quad E_{\text{spec}} := \left\| \sum_{p \leq \kappa} c_p a_p \right\|^2 = \|\Phi c\|^2,$$

where $D := \text{diag}(\|a_p\|^2)_{p \in S(\kappa)}$. The energy asymmetry is

$$\Delta := E_{\text{str}} - E_{\text{spec}} \geq 0 \quad (\text{numerically}), \quad \eta_{\text{orig}} := \frac{\Delta}{E_{\text{str}}} = 1 - \frac{c^T G^{\text{un}} c}{E_{\text{str}}}.$$

2.2 Definition of the Tehrani Operator $\tilde{T}$

Definition 2.7 (Tehrani operator).

$$\tilde{T} := \Phi \circ \Phi^*: H_{\text{null}} \rightarrow H_{\text{null}}.$$

Matrix representation. In the standard basis $\{f_k\}_{k=1}^N$ of H_{null} :

$$\tilde{T}_{kl} = \sum_{p \leq \kappa} (a_p)_k (a_p)_l = \sum_{p \leq \kappa} e^{-\varepsilon^2 \gamma_k^2 / 2} \sin(\gamma_k \log p) \cdot e^{-\varepsilon^2 \gamma_l^2 / 2} \sin(\gamma_l \log p).$$

Proposition 2.8 (Properties of $\tilde{T}$). *The operator $\tilde{T}$ satisfies:*

- (i) **Self-adjointness:** $\tilde{T}^* = (\Phi \Phi^*)^* = \Phi \Phi^* = \tilde{T}$. ✓
- (ii) **Positive semi-definiteness:** For all $u \in H_{\text{null}}$, $\langle \tilde{T}u, u \rangle = \|\Phi^*u\|^2 \geq 0$. ✓
- (iii) **Rank:** $\text{rank}(\tilde{T}) = \text{rank}(\Phi) = \pi(\kappa)$. ✓
- (iv) **Explicit construction:** $\tilde{T}$ is built directly from prime resonance vectors $\{a_p\}$. The zero ordinates γ_k are numerical inputs to the resonance vectors and hence to $\tilde{T}$. Non-circularity means: no RH, GUE, or HP assumption is used; the γ_k are external inputs, not conclusions of the model.

Proof. Parts (i) and (ii) are immediate from $\tilde{T} = \Phi \Phi^*$ using standard properties of adjoints. Part (iii) follows from $\text{rank}(AB) \leq \min(\text{rank}(A), \text{rank}(B))$ and $\pi(\kappa) \leq N$ in all normative parameter choices. $\square$

2.3 Algebraic Spectral Identity

Theorem 2.9 (Spectral identity; proved).

$$\sigma(T) \setminus \{0\} = \sigma(\tilde{T}) \setminus \{0\}.$$

The shared nonzero eigenvalues satisfy:

$$\max_{j=1}^{\pi(\kappa)} |\lambda_j(T) - \mu_j(\tilde{T})| < 10^{-15} \quad (\text{machine precision, verified numerically}).$$

Proof. Classical result (see e.g. [6], Appendix A, or any standard linear algebra text): for any A, B , $\sigma(AB) \setminus \{0\} = \sigma(BA) \setminus \{0\}$. Applied with $A = \Phi^*$, $B = \Phi$: $T = \Phi^* \Phi$ and $\tilde{T} = \Phi \Phi^*$, so $\sigma(T) \setminus \{0\} = \sigma(\tilde{T}) \setminus \{0\}$. (For finite-dimensional matrices, Sylvester's determinant identity: $\det(\lambda I - AB) = \lambda^{n-m} \det(\lambda I - BA)$ for $A \in \mathbb{R}^{m \times n}$, $B \in \mathbb{R}^{n \times m}$.) $\square$

Remark. The identity $\lambda_j(T) = \mu_j(\tilde{T})$ is an algebraic theorem — it holds in exact arithmetic for any choice of κ , ε , and N . The eigenvalues $\lambda_j = \mu_j$ are not the zero ordinates γ_k ; they are constructive spectral quantities arising from Φ .

2.4 $\tilde{T}$ is NOT a Hilbert–Pólya Operator

This section records a negative result.

Proposition 2.10 (Numerical negative result). *A Hilbert–Pólya operator $\mathcal{H}$ would satisfy $\mathcal{H}\psi_k = \gamma_k \psi_k$ for all k , i.e., the zero ordinates γ_k appear directly as eigenvalues. $\tilde{T}$ does **not** have this property:*

- (a) [Numerical] The eigenvalues of $\tilde{T}$ satisfy $\mu_j \sim C_T/\gamma_{k(j)}$ — they are inversely proportional to zero ordinates (§3.1).
- (b) [Numerical] The zero ordinates γ_k appear as localization centers of eigenvectors v_j (§3.2), not as eigenvalues.
- (c) [Numerical] The inverse operator $W_1 := C_T \cdot \tilde{T}^+$ has eigenvalues $\omega_j = C_T/\mu_j \approx \gamma_{k(j)}$ for small j , but the correlation $r_2 := \text{corr}(\omega_j, \gamma_{k(j)})$ falls from 0.502 (at $\kappa = 23$) to 0.160 (at $\kappa = 503$). W_1 is therefore **not** an asymptotic HP-operator.

The fundamental distinction:

$$\text{HP-operator: eigenvalues} = \gamma_k. \quad \tilde{T}: \text{eigenvector localization centers} = \gamma_k.$$

$\tilde{T}$ is a **resonance operator**: it encodes the γ_k as mode centers, not as spectral values.

Proof of (a)(b)(c). Numerical; see §3.1, §3.2, §3.4 for detailed data. $\square$

Remark (Structural obstruction). The Gaussian damping $e^{-\varepsilon^2 \gamma_k^2/2}$ in a_p causes μ_j to decay super-exponentially (exponent $\alpha \approx -4.88$), far faster than the $1/\gamma$ -law (exponent -0.56). Inverting $\tilde{T}$ to form W_1 catastrophically amplifies the smallest eigenvalues. For $j = 16$ at $\kappa = 53$: $\omega_{16} \approx 7 \times 10^6$ against $\gamma_{16} \approx 67$. This is the structural reason W_1 fails as an HP-operator (see §3.4 and §6).

3 Spectral Analysis of $\tilde{T}$

3.1 Eigenvalue Structure: $\mu_j \sim C_T/\gamma_{k(j)}$

Numerical evidence (Eigenvalue–Zero Ordinate Correlation).

Let $k_{\text{dom}}(j) := \arg \max_k |(v_j)_k|$ be the dominant zero index for eigenvector v_j . Then:

j	$\mu_j(\tilde{T})$	k_{dom}	$\gamma_{k_{\text{dom}}}$	Localization
1	4.219803	1	14.1347	0.985
2	2.565694	3	25.0109	0.741
3	1.744649	2	21.0220	0.745
4	0.986492	4	30.4249	0.854
5	0.314199	5	32.9351	0.794
6	0.217900	6	37.5862	0.885

The OLS correlation between μ_j and $1/\gamma_{k(j)}$ is

$$r_1 = \text{corr}(\mu_j, 1/\gamma_{k(j)}) = 0.950 \quad (\kappa = 53).$$

For $\kappa = 503$: $r_1 = 0.859$ (weakening as κ grows, due to Gaussian damping).

Remark. The dominant index satisfies $k_{\text{dom}}(1) = 1$ for all tested $\kappa \in \{23, 53, 101, 199, 503, 1009\}$: the first eigenvector of $\tilde{T}$ is always associated with $\gamma_1 = 14.1347$.

Remark (Cauchy–Schwarz warning). Note that $r_1 = 0.950$ coexists with $r_2 = 0.502$ (§3.4). These measure different things: r_1 tests monotone ordering; r_2 tests value-level approximation. The distinction between ordering and value accuracy is crucial for the HP assessment in §2.4.

3.2 Eigenvector Localization

Numerical evidence (Eigenvector Localization).

For each eigenvector v_j of $\tilde{T}$ (normalized: $\|v_j\| = 1$), the localization is measured by $L_j := |(v_j)_{k_{\text{dom}}(j)}|$. Observed range: $L_j \in [0.45, 0.99]$ for $j = 1, \dots, 14$.

For comparison, a uniformly random unit vector in $\mathbb{R}^{100}$ has typical entry magnitude $1/\sqrt{100} = 0.10$. The observed localization is 4.5–9.9 times the random baseline. The localization structure suggests that $\tilde{T}$ encodes the ordering of zero ordinates in a distributional sense: each mode v_j is concentrated near the zero $\gamma_{k(j)}$ in the zero-space H_{null} . This is the precise content of calling $\tilde{T}$ a resonance operator.

3.3 The Two Arithmetic Constants C_η and C_T

Definition 3.1 (Arithmetic constants).

$$C_\eta := \lim_{\kappa \rightarrow \infty} \frac{\lambda_{\max}(T_{\text{ren}})}{\pi(\kappa)}, \quad C_\eta \approx 0.39,$$

where $T_{\text{ren}} := D^{-1/2}TD^{-1/2}$.

$$C_T(\kappa) := \text{OLS-slope of } \mu_j \sim C_T/\gamma_{k(j)}, \quad C_T/\pi(\kappa) \in \{2.49, 2.21, 2.17, 1.94, 1.89\}$$

for $\kappa \in \{23, 53, 101, 199, 503\}$ (weakly decreasing).

$$C_\eta \neq C_T.$$

These are two distinct arithmetic quantities; always use subscripts.

The constant $C_\eta \approx 0.39$ is κ -independent; C_T grows with κ (numerically observed to grow approximately as $2\pi(\kappa)$).

Numerical evidence ($\lambda_{\max, \text{ren}}$ growth).

κ	$\pi(\kappa)$	$\lambda_{\max, \text{ren}}$	Ratio
23	9	3.59	0.399
53	16	6.38	0.399
101	26	10.35	0.398
199	46	17.76	0.386
503	96	37.38	0.389
1009	169	65.16	0.386

Variance of the ratio: $\sigma^2 < 0.0003$ — extremely stable. The universal bound $\lambda_{\max, \text{ren}} < 1$ (an earlier conjecture in this series, now falsified) is numerically falsified: $\lambda_{\max, \text{ren}} \rightarrow \infty$.

Remark (Analytical status of C_η). *The value $C_\eta \approx 0.39$ is not analytically explained. A candidate formula $C_\eta = 1/I_N(\varepsilon)$ is **falsified**: it gives $1/I_{100}(0.05) \approx 0.73 \neq 0.39$. The correct representation involves a harmonic mean (analytically open; see §6 and §6).*

3.4 The Inverse Operator $W_1 = C_T \cdot \tilde{T}^+$

Definition 3.2 (Inverse operator). $W_1 := C_T \cdot \tilde{T}^+$, where $\tilde{T}^+$ denotes the Moore–Penrose pseudoinverse of $\tilde{T}$ and C_T is the OLS constant from Definition 3.1.

Proposition 3.3 (Self-adjointness of W_1 ; proved). W_1 is self-adjoint.

Proof. $\tilde{T}$ is self-adjoint (Proposition 2.8(i)). For any self-adjoint operator, the Moore–Penrose pseudoinverse is also self-adjoint (standard result in linear algebra). C_T is a real scalar. Therefore $W_1 = C_T \cdot \tilde{T}^+$ is self-adjoint. $\square$

Numerically: $\|W_1 - W_1^T\|_\infty = 3.1 \times 10^{-4}$ (machine precision effects from eigendecomposition).

Proposition 3.4 (Non-circularity status of W_1 ; proved). $W_1 = C_T \cdot \tilde{T}^+$ does not assume RH or the Hilbert–Pólya postulate. However, its normalization constant C_T is the OLS slope in $\mu_j \sim C_T/\gamma_{k(j)}$, which is calibrated against the zero ordinates. W_1 should therefore not be presented as fully input-free. $\tilde{T}$ is explicit and deterministic; the γ_k are external numerical inputs, not conclusions of the model. Proof. $\tilde{T}$ is constructed from prime resonance vectors (Definition 2.7), using γ_k as numerical inputs $\checkmark$. $\tilde{T}^+$ is the Moore–Penrose pseudoinverse — no postulate $\checkmark$. The scalar C_T is computed via OLS regression of μ_j against $1/\gamma_{k(j)}$: this uses γ_k as calibration values, but does not assume RH or HP. $\square$

Numerical evidence (W_1 HP-correlation failure).

κ	$\pi(\kappa)$	$r_2 = \text{corr}(\omega_j, \gamma_{k(j)})$
23	9	0.502
53	16	0.436
101	26	0.321
199	46	0.226
503	96	0.160

The decreasing trend confirms: W_1 is not an asymptotic HP-operator. The structural cause is Gaussian damping: $e^{-\varepsilon^2 \gamma_k^2/2}$ in Φ causes superexponential decay of μ_j (observed exponent $\alpha \approx -4.88$), far exceeding the $1/\gamma$ growth pattern required for HP. Pseudoinversion catastrophically amplifies tail modes.

Summary for § 3.4: W_1 is self-adjoint (proved). Its construction does not assume RH or the HP postulate, but C_T is calibrated via OLS on the zero ordinates — W_1 is therefore not fully input-free (Proposition 3.4). W_1 is the natural HP-candidate but fails numerically as κ grows. This rules out this particular pseudoinverse construction as an HP candidate. This does not preclude other spectral functions $f(\tilde{T})$ from being valid HP-candidates; see § 6.

4 The Monotonicity Mechanism

The strategy for bounding η_{orig} from below rests on the interplay of three structural features: the monotone weight sequence (c_p) , the oscillatory mode structure $(\sin(\gamma_k \log p))$, and Abel summation.

4.1 Monotone weight structure

Lemma 4.1 (Monotone weight structure; proved for $p \geq 11$, $K = 1$). *For $K = 1$ (leading term of $V_p(\sigma)$) and $p \geq 11$:*

$$c_p \approx \frac{4 \log p}{\sqrt{p}} \quad \text{is strictly monotone decreasing in } p.$$

Proof. Differentiate with respect to p (treating p as continuous): $\frac{d}{dp}[\log p / \sqrt{p}] = (1/2 - \log p)/(p\sqrt{p}) < 0$ for $p > e^{1/2} \approx 1.65$. This holds for all primes $p \geq 2$. $\square$

Remark (Status for $K > 1$). When $K > 1$ (i.e., $p^2 \leq \kappa$), the weight formula $V_p(\sigma, K)$ is more complex. The monotonicity $V_p(\sigma, K+1) \geq V_p(\sigma, K)$ is numerically supported but not formally proved. The general monotonicity of c_p for all p with $K > 1$ is therefore OPEN.

4.2 Oscillatory mode structure

Lemma 4.2 (Oscillatory mode structure; partial). (M2a; proved). *The function $p \mapsto \sin(\gamma_k \log p)$ changes sign in the intervals*

$$I_m := [e^{m\pi/\gamma_k}, e^{(m+1)\pi/\gamma_k}], \quad m = 1, 2, 3, \dots$$

Proof. Direct from $\sin(x) = 0$ at $x = m\pi$: the argument $\gamma_k \log p$ crosses $m\pi$ at $p = e^{m\pi/\gamma_k}$. $\square$

(M2b; open). By the Prime Number Theorem, each interval I_m contains at least one prime $p \leq \kappa$, for m in a range depending on γ_k and κ . Full quantitative control for all m is OPEN.

4.3 Abel Summation Principle

Lemma 4.3 (Abel summation principle; proved). *Let $P \geq 1$. Let $(f_1, \dots, f_P)$ be monotone non-increasing and non-negative. Let $(g_1, \dots, g_P)$ be any sequence with partial sums bounded by M : $|G_m| := |\sum_{i=1}^m g_i| \leq M$ for all m . Then:*

$$\left| \sum_{i=1}^P f_i g_i \right| \leq M \cdot f_1.$$

Proof. Apply the Abel partial summation formula: $\sum_{i=1}^P f_i g_i = f_P G_P - \sum_{i=1}^{P-1} (f_{i+1} - f_i) G_i$. Taking absolute values:

$$\begin{aligned} \left| \sum_{i=1}^P f_i g_i \right| &\leq |f_P| \cdot |G_P| + \sum_{i=1}^{P-1} |f_{i+1} - f_i| \cdot |G_i| \\ &\leq M f_P + M \sum_{i=1}^{P-1} (f_i - f_{i+1}) \quad [f \text{ decreasing}] \\ &= M f_P + M(f_1 - f_P) = M f_1. \quad \square \end{aligned}$$

Remark (Application context for Paper 4). *With $f_i = (D^{-1/2} \tilde{c})_{p_i} \propto c_{p_i} / \|a_{p_i}\|$ (monotone decreasing — Lemma 4.1) and $g_i = (u_j)_{p_i}$ (oscillatory eigenvector entries — Lemma 4.2), Lemma 4.3 bounds the projection $|\langle u_j, D^{-1/2} \tilde{c} \rangle| \leq M_{k(j)}(\kappa) \cdot c_{p_1} / \|a_{p_1}\|$, provided the partial sums of u_j -entries are bounded by $M_{k(j)}(\kappa)$. [The full reduction is structural, not a direct consequence of Lemma 4.3 alone.]*

4.4 Prime Exponential Sum Bound

Theorem 4.4 (Prime exponential sum bound; proved). *For fixed $\gamma > 0$ and $\kappa \geq 2$:*

$$M_k(\kappa) := \max_{\substack{x \leq \kappa \\ p \leq x}} \left| \sum \sin(\gamma_k \log p) \right| = O\left(\frac{\pi(\kappa)}{\gamma_k}\right).$$

Explicitly:

$$M_k(\kappa) \leq \left(\frac{2}{\gamma_k} + o\left(\frac{1}{\gamma_k}\right) \right) \pi(\kappa), \quad \kappa \geq 2.$$

Proof. We establish three auxiliary lemmas.

Lemma 4.5 (Abel–Stieltjes representation; proved). *For $\gamma > 0$ and $\kappa \geq 2$:*

$$\sum_{p \leq \kappa} p^{i\gamma} = A_0(\gamma) + J_\gamma(\kappa) + R_\gamma(\kappa),$$

where $|A_0(\gamma)| \leq 0.1$; $J_\gamma(\kappa) := \int_2^\kappa x^{i\gamma} / \log x \, dx$ (main term); $R_\gamma(\kappa) := \kappa^{i\gamma} E(\kappa) - i\gamma \int_2^\kappa E(x) x^{i\gamma-1} dx$, with $E(x) = O(xe^{-c\sqrt{\log x}})$ (error term). No RH is used; PNT provides the $E(x)$ bound. $\square$

Lemma 4.6 (Main term bound; proved).

$$|J_\gamma(\kappa)| \leq \left(\frac{2}{\gamma} + o\left(\frac{1}{\gamma}\right) \right) \pi(\kappa).$$

Proof. The dominant term satisfies $|\kappa^{1+i\gamma}/((1+i\gamma)\log \kappa)| = \kappa/(\sqrt{1+\gamma^2} \cdot \log \kappa) \leq \kappa/(\gamma \log \kappa)$. Since $\pi(\kappa) \sim \kappa/\log \kappa$ (PNT): $|J_\gamma(\kappa)| \leq (2/\gamma)\pi(\kappa)(1 + O(1/\log \kappa))$. $\square$

Lemma 4.7 (Error term control; proved). For fixed $\gamma > 0$ and $\kappa \rightarrow \infty$: $|R_\gamma(\kappa)| = o(\pi(\kappa))$.

Proof. PNT (Hadamard/de la Vallée-Poussin): $|E(x)| = O(xe^{-c\sqrt{\log x}})$. Substitution $u = \sqrt{\log x}$ yields $\int_2^\kappa e^{-c\sqrt{\log x}} dx = O(\sqrt{\log \kappa} \cdot \kappa \cdot e^{-c\sqrt{\log \kappa}}) = o(\pi(\kappa))$, since $\sqrt{\log \kappa} \cdot e^{-c\sqrt{\log \kappa}} = o(1/\log \kappa)$. Combined: $|R_\gamma(\kappa)| = o(\pi(\kappa))$ for fixed γ . $\square$

Proof of Theorem 4.4: The imaginary part of Lemma 4.5 gives $A_k(\kappa) = \text{Im}[A_0] + \text{Im}[J_{\gamma_k}] + \text{Im}[R_{\gamma_k}]$. Applying Lemmas 4.6 and 4.7:

$$|A_k(\kappa)| \leq 0.1 + \frac{2}{\gamma_k} \pi(\kappa)(1 + o(1)) + o(\pi(\kappa)) = O\left(\frac{\pi(\kappa)}{\gamma_k}\right).$$

The same bound holds for all partial sums (Remark 4.5), so the maximum $M_k(\kappa)$ satisfies the same bound. $\square$

Corollary 4.8. For $k \rightarrow \infty$ (with $\gamma_k \rightarrow \infty$): $M_k(\kappa)/\pi(\kappa) \rightarrow 0$ uniformly in κ .

Remark (Goldston–Gonek confirmation [5]). An independent formula of Goldston–Gonek yields the same $O(\pi(\kappa)/\gamma_k)$ bound via mean-value methods. This provides an independent check. It provides no improvement for fixed k , confirming that the bottleneck is the joint equidistribution Assumption.

Remark (Methodological note). The estimate $M_k(\kappa) = O(\pi(\kappa)/\gamma_k)$ is proved by standard PNT/Abel-summation technology; the analytic method has precedents in Turán’s program on approximative Dirichlet polynomials [8]. What is specific to the present paper is not the bare oscillatory-sum bound, but the way this bound enters the conditional energy-asymmetry argument of § 5.

Remark (What Theorem EXPLICIT does NOT prove). Theorem 4.4 does not prove $M_k(\kappa) = o(\pi(\kappa))$ for fixed k as $\kappa \rightarrow \infty$. This stronger statement is equivalent (via Weyl’s criterion) to equidistribution of $\{\gamma_k \log p \bmod 2\pi\}$, which is the genuinely open Assumption (§ 6).

5 Results on $\eta_{\text{orig}} > 0$

5.1 Numerical Confirmation

Numerical evidence ($\eta_{\text{orig}} > 0$ for $\kappa \leq 1009$).

Let $\tilde{c} = D^{1/2}c/\sqrt{E_{\text{str}}}$ be the normalized canonical weight vector. The Rayleigh quotient identity

$$\langle T_{\text{ren}} \tilde{c}, \tilde{c} \rangle = 1 - \eta_{\text{orig}}$$

holds to machine precision ($< 2 \times 10^{-16}$), where $T_{\text{ren}} := D^{-1/2} T D^{-1/2}$.

κ	$\pi(\kappa)$	$\langle T_{\text{ren}} \tilde{c}, \tilde{c} \rangle$	$< 1?$	η_{orig}
23	9	0.18849	✓	0.81151
53	16	0.33073	✓	0.66927
101	26	0.22803	✓	0.77197
199	46	0.21740	✓	0.78260
503	96	0.21788	✓	0.78212
1009	169	0.18002	✓	0.81998

$\eta_{\text{orig}} > 0$ for all tested $\kappa \leq 1009$ ✓.

Collective mechanism. The dominant contribution to $\langle T_{\text{ren}} \tilde{c}, \tilde{c} \rangle = \sum_j B_j$ comes from mode $j = 4$ ($B_4 = 0.193$), not from the dominant eigenmode $j = 1$ ($B_1 = 0.040$). Approximately 92% cancellation occurs at $j = 1$ ($\kappa = 53$). The effect is distributed across ~ 8 eigenmodes — $\eta_{\text{orig}} > 0$ is a **collective spectral effect**, not a simple u_1 -orthogonality principle.

5.2 Δ -Decomposition and Burst Stability

Definition 5.1 (Δ -decomposition). With $S(r) := \sum_{k=1}^N e^{-\varepsilon^2 \gamma_k^2} \cos(\gamma_k \log r)$:

$$\Delta = \Delta_{\text{Burst}} + \Delta_{\text{Cross}} + \Delta_{\text{Stream}},$$

where

$$\Delta_{\text{Burst}} = \sum_{p < q \leq 7} c_p c_q [S(pq) - S(p/q)], \quad \Delta_{\text{Cross}} = \sum_{p \leq 7 < q \leq \kappa} c_p c_q [S(pq) - S(p/q)],$$

$$\Delta_{\text{Stream}} = \sum_{7 < p < q \leq \kappa} c_p c_q [S(pq) - S(p/q)].$$

The factorization $S(pq) - S(p/q) = -2 \sum_k e^{-\varepsilon^2 \gamma_k^2} \sin(\gamma_k \log p) \sin(\gamma_k \log q)$ follows from $\cos(A + B) - \cos(A - B) = -2 \sin A \sin B$. [proved]

Numerical evidence (Δ_{Burst} stability).

Pair	$S(pq)$	$S(p/q)$	$c_p c_q$	Contribution
(2, 3)	0.9536	0.2653	≈ 2.408	+1.657
(2, 5)	0.4110	0.6555	≈ 1.833	−0.448
(2, 7)	0.5794	0.5812	≈ 1.501	−0.003
(3, 5)	0.9440	0.3686	≈ 1.531	+0.881
(3, 7)	0.7193	0.5571	≈ 1.254	+0.203
(5, 7)	0.9668	0.1135	≈ 0.954	+0.814
Total				$\approx +\mathbf{3.11}$

$\Delta_{\text{Burst}} \approx 3.11 > 0$, and it is κ -**invariant** for all tested $\kappa > 7$. The sufficient condition for $\Delta > 0$ is:

$$|\Delta_{\text{Cross}} + \Delta_{\text{Stream}}| < \Delta_{\text{Burst}} \approx 3.11.$$

This is numerically satisfied for all $\kappa \leq 1009$; analytic proof is OPEN.

5.3 Conditional Theorem

Theorem 5.2 (Conditional and numerical results for $\eta_{\text{orig}} > 0$). ***Assumption** (Weyl equidistribution). The sequences $\{(\gamma_{k_1} \log p, \gamma_{k_2} \log p) \bmod 2\pi\}_{p \leq \kappa, \kappa \rightarrow \infty}$ are jointly equidistributed in $[0, 2\pi)^2$ for all $k_1 \neq k_2$ with $k_1, k_2 \leq N$. This concerns $\text{Re}(s) = 1$, distinct from RH which concerns $\text{Re}(s) = \frac{1}{2}$.*

Under this Assumption:

- (i) [Conditional] *The sequence $\{\gamma_k \log p \bmod 2\pi\}_{p \leq \kappa, \kappa \rightarrow \infty}$ is equidistributed modulo 2π for each fixed $k \leq N$.*
- (ii) [Conditional] *The normalized unweighted cross averages vanish: $\frac{1}{\pi(\kappa)} \sum_{p \leq \kappa} \sin(\gamma_{k_1} \log p) \sin(\gamma_{k_2} \log p) \rightarrow 0$ for all $k_1 \neq k_2$.*
- (iii) [Numerical — no assumption needed] $\Delta(\kappa) \geq \Delta_{\text{Burst}} \approx 3.11 > 0$ for all tested $\kappa > 7$, κ -invariant by construction.
- (iv) [Numerical] $\eta_{\text{orig}}(\kappa) \rightarrow \eta_{\infty} \approx 0.81 > 0$, with unconditional lower bound $\eta_{\infty} \geq \Delta_{\text{Burst}}/E_{\text{str}}(\infty) \approx 0.51 > 0$.

Proof sketch. (i) follows from the Assumption by Weyl's criterion (cf. [6], §21.1); the required non-vanishing $\zeta(1 + in\gamma_k) \neq 0$ is unconditional (Hadamard 1896). (ii) follows from the Assumption via $\sin A \sin B = \frac{1}{2}[\cos(A - B) - \cos(A + B)]$. (iii) and (iv) are numerical observations; no assumption is used. $\square$

Remark (Correction to earlier formulation). *The earlier claim “ $\Delta(\kappa) \rightarrow \Delta_{\text{Burst}}$ ” was incorrect: $\Delta_{\text{Burst}} \approx 3.11$ is a κ -invariant lower bound, not the limit. Numerically, $\Delta_{\infty} \approx 4.9 > \Delta_{\text{Burst}}$.*

Remark (Connection to RH). *The implication chain is:*

$$\text{RH} \longleftarrow [\text{transfer open}] \longleftarrow \eta_{\text{orig}} > 0 \longleftarrow \text{Theorem 5.2} \longleftarrow \text{Assumption}.$$

No direct connection between $\eta_{\text{orig}} > 0$ and RH is established in this paper. The missing step is a quantitative transfer from $\sigma = \frac{1}{2}$ to $\sigma \neq \frac{1}{2}$ on the curvature level — entirely outside the scope of this paper.

6 Open Problems

Open Problem 6.1 (Weyl equidistribution). *Prove or disprove: the sequences $\{(\gamma_{k_1} \log p, \gamma_{k_2} \log p) \bmod 2\pi\}_{p \leq \kappa, \kappa \rightarrow \infty}$ are jointly equidistributed in $[0, 2\pi)^2$ for all $k_1 \neq k_2$, $k_1, k_2 \leq N$.*

The non-vanishing $\zeta(1 + in\gamma_k) \neq 0$ required by Weyl's criterion is unconditional (Hadamard 1896). The equidistribution itself is genuinely open.

Possible approaches: (A) Hadamard product estimates near $s = 1 + in\gamma_k$. (B) Linear independence of $\{\gamma_k \log p\}$ over $\mathbb{Q}$. (C) Numerical verification for small n, k .

Open Problem 6.2 (Analytic positivity). Prove $\langle T_{\text{ren}} \tilde{c}, \tilde{c} \rangle < 1$ for the canonical weight system without the equidistribution assumption of Theorem 5.2.

Current best: $M_k(\kappa) = O(\pi(\kappa)/\gamma_k)$ (Theorem 4.4) bounds individual projections but is insufficient for the full sum $\sum_j \lambda_j |\langle u_j, D^{-1/2} \tilde{c} \rangle|^2 < E_{\text{str}}$.

Open Problem 6.3 (Circularity-free spectral function). Does any circularity-free spectral function $f(\tilde{T})$ satisfy $\text{Spec}(f(\tilde{T})) \rightarrow \{\gamma_k\}$ as $\kappa \rightarrow \infty$?

The natural candidate $W_1 = C_T \cdot \tilde{T}^+$ fails: $r_2 \rightarrow 0.16$ (§3.4). The structural obstruction is Gaussian damping.

A possible modification: replace $e^{-\varepsilon^2 \gamma_k^2/2}$ by $e^{-\varepsilon \gamma_k}$ (slower decay) in Φ .

Open Problem 6.4 (Arithmetic constant C_η). Determine the analytic formula for $C_\eta = \lim_{\kappa \rightarrow \infty} \lambda_{\text{max,ren}}/\pi(\kappa)$. The candidate $C_\eta = 1/I_N(\varepsilon)$ is falsified. The correct representation involves a harmonic mean (analytically open).

Open Problem 6.5 (Asymptotics of C_T). Does $C_T(\kappa)/\pi(\kappa)$ converge as $\kappa \rightarrow \infty$? Numerically: values 2.49, 2.21, 2.17, 1.94, 1.89 (decreasing) — the limit, if it exists, is analytically undetermined.

Open Problem 6.6 (Monotone bound). Is $\eta_{\text{orig}} > 0$ a consequence of the Prime Number Theorem alone, independent of any RH-related hypothesis?

Evidence: $\Delta_{\text{Burst}} \approx 3.11 > 0$ depends only on primes $p \leq 7$ and is κ -invariant. Analytically OPEN.

Remark (Connection to RH). The implication chain:

$$\text{RH} \leftarrow [\text{transfer open}] \leftarrow \eta_{\text{orig}} > 0 \leftarrow \text{Theorem 5.2} \leftarrow \zeta(1 + in\gamma_k) \neq 0.$$

No direct connection between $\eta_{\text{orig}} > 0$ and RH is established in this paper. The missing step is a quantitative transfer from $\sigma = \frac{1}{2}$ to $\sigma \neq \frac{1}{2}$ on the curvature level — a step entirely outside the scope of this paper and the subject of future work.

7 Non-Circularity

We document explicitly that the results of this paper do not rely on the objects they are directed toward. The following holds unconditionally throughout §§2–5.

No RH as input. At no point is the Riemann Hypothesis assumed. The zero ordinates γ_k enter the construction of Φ as numerically specified inputs, and no property of their distribution

requiring the Riemann Hypothesis is assumed or used. In Theorem 5.3, the Assumption is explicitly labelled CONDITIONAL; it concerns $\operatorname{Re}(s) = 1$, distinct from RH which concerns $\operatorname{Re}(s) = \frac{1}{2}$. Its logical relationship to RH is not analysed in this paper.

No GUE, Montgomery, or Hilbert–Pólya hypothesis. The Tehrani operator $\tilde{T} = \Phi \circ \Phi^*$ is a deterministic, explicitly computable finite-dimensional operator, constructed from the prime resonance vectors $\{a_p\}_{p \leq \kappa}$. It is not a random matrix, not a statistical ensemble, and not a Hilbert–Pólya operator. No GUE conjecture, no Montgomery pair correlation conjecture, and no random-matrix-theoretic assumption is used or implied anywhere in §§ 2–5. The structural similarity between eigenvector localization in $\tilde{T}$ and GUE level statistics is a numerical observation, not a postulate.

Status separation maintained. The zero ordinates γ_k appear as inputs to the construction of Φ , never as conclusions. The eigenvalues μ_j of $\tilde{T}$ are derived from Φ and are not identified with the γ_k .

Note on W_1 . The inverse operator $W_1 = C_T \cdot \tilde{T}^+$ does not assume RH or the Hilbert–Pólya postulate; however, its normalization constant C_T is calibrated against the zero ordinates via OLS (§ 3.3). W_1 is therefore not fully input-free. $\tilde{T}$ itself is deterministic and explicitly computable.

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Call for Feedback

This paper is part of an ongoing series toward the Riemann Hypothesis. All papers are freely available on Zenodo under CC BY 4.0 (<https://zenodo.org/search?q=Ulrich+Tehrani>), and the accompanying code is hosted on GitHub (<https://github.com/utehrani/analysislab-nt>).

The author welcomes critical feedback, corrections, and collaboration. In particular: analytical approaches to the Weyl equidistribution of $\{\gamma_k \log p \bmod 2\pi\}$ (Open Problem in § 6), connections to existing results on exponential sums over primes and zeros of $\zeta(s)$, and any errors or gaps in the present arguments.

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